DAY SHIFT
AGE-RESPECTFUL UPPER-LIMB REHABILITATION GAMES

CS 435 – HCI Group Course Project

Assignment 3: High-Fidelity Prototyping and User Evaluation

Group 1 · Demo video: Demo_ Video.mp4
Repository: github.com/Jain-Laksh/HCI-Upper-Limb-Rehabilitation-Gamified-Platform

Akshat A. Nishad Rudra P. Singh
Laksh Jain Harshil H Shah

Indian Institute of Technology Gandhinagar

Instructor: Dr. Yogesh K. Meena

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Abstract

Day Shift is a home tele-rehabilitation game for older adults. The user works a day as a **house painter**: each upper-limb exercise is masking tape on a wall, the hand is the brush, and paint appears only while the movement follows the tape. The high-fidelity prototype is a web app that runs entirely in the browser with an ordinary webcam. Exercises are stored in the user's own body units, so one reference movement fits every body; depth toward the camera is recovered from calibrated bone lengths; and a therapist can turn a phone video into a new exercise. The interface is designed for older adults: large readable text, high contrast, dwell selection with the free hand, spoken guidance with captions, voice commands and Hindi. This report describes the design, the system, the compromises made and the evaluation plan.

1 Introduction

Older adults recovering from stroke or injury are asked to repeat arm exercises at home, which is dull and easy to abandon. In Assignment 1 we found that most rehabilitation games use cartoon visuals and child-like rewards that older users can find patronising. Assignment 2 proposed *vocational role-play*: familiar adult work gives each movement a purpose. Assignment 3 builds a working **vertical slice** of that idea: one trade, the House Painter, implemented end to end, from a therapist's reference movement to a pay slip. Other trades are visible but not playable.

2 Conceptual Design

2.1 Metaphor

The conceptual model is *a working day as a house painter* (Table 1). Users can reason with knowledge they already have: tape marks where paint goes, a neat painter stays on the tape, and a day has a start, a lunch break and an end. Painting replaced Assignment 2's gardening because it maps directly onto continuous arm paths and shows progress as a wall filling with colour.

Game element	Stands for	Design intent
Masking tape	Exercise path	Familiar "paint here" signifier; tape width is the tolerance
Brush	User's hand	Direct mapping: the brush moves as the hand moves
Paint appearing	Being on the path	Continuous feedback without numbers
Stroke / wall / job	Repetition / set / session	Natural hierarchy: "two more strokes on this wall"
Clock 9:00–17:00, lunch	Session progress, rest	Moves with work done, not real time: no time pressure
Wage and pay slip	Effort and summary	Adult motivation; never deducted; clinical numbers kept separate

Table 1: Conceptual model.

2.2 Design principles

Table 2 shows how Norman's principles appear in the interface. We also apply Fitts' law (large targets near the hand that uses them), Hick's law (at most six choices per screen), WCAG 2.2 contrast, and *dignity*: adult wording, no childish praise and no penalties.

Principle	In the prototype
Discoverability	Few large, labelled buttons; Help and Back on every screen; "Raise your hand to point" hint
Feedback	Paint under the brush, glowing tape, filling dwell ring, sounds, spoken and captioned messages
Conceptual model	Painter, shift and wage metaphor; each wall opens with an animated demo of the stroke
Affordances	Raised, bordered buttons at least 72 px high; the brush drawn as a round tool head
Signifiers	Dashed tape with direction arrows, green start circle, dashed end circle
Mapping	Mirrored camera image; tape anchored to the user's own shoulder; clock fills left to right
Constraints	Paint only inside the tape; a stroke must begin on the start circle; "End shift" asks to confirm

Table 2: Norman's design principles in the interface.

Changes from the low-fidelity design. Gardening became painting; the "two photos of the arm" became a live three-second measurement; fist selection became dwell, because grip is often impaired; the real-time 9-to-5 timer became a work-driven clock; and seated play became the default.

3 System Design and Architecture

3.1 Architecture and technology

Day Shift is a single-page web app with no server (Figure 1). All processing, including pose estimation and video analysis, happens in the browser, so camera images never leave the device and the app can be hosted as static files. Table 3 lists the stack.

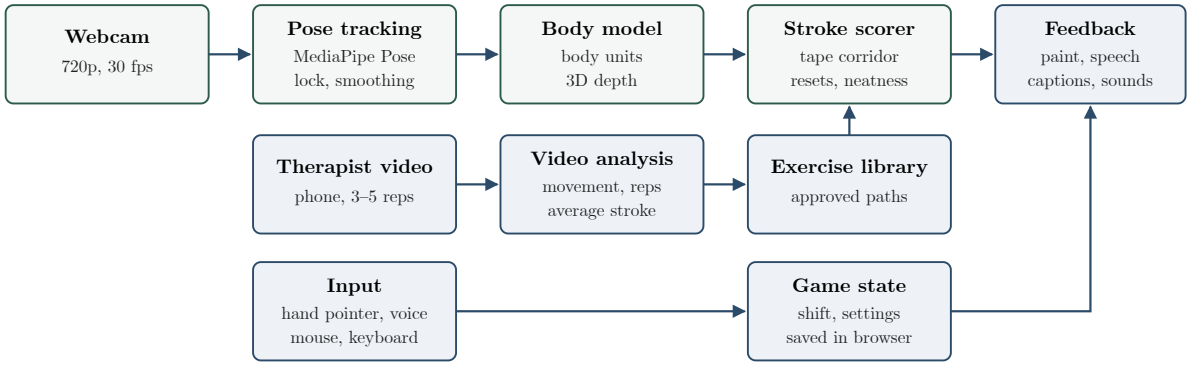


Figure 1: Architecture. Top: the live loop for every camera frame. Middle: turning a therapist's video into an exercise (it reuses pose tracking and the body model). Bottom: menus and saved state.

Concern	Technology	Why
Application	React 18, TypeScript, Vite	Component screens; type-checked geometry code
Pose	MediaPipe PoseLandmarker (BlazePose), full or lite model, bundled	Free, GPU in the browser, gives 3D world landmarks
State	zustand, saved to localStorage	Settings, progress and an unfinished shift survive a reload
Speech	Web Speech API: synthesis (out), recognition (in)	No downloads or keys; English (India) and Hindi voices
Sound, motion	Web Audio tones; Framer Motion	Full control of pitch and volume; motion can be turned off
Type, icons	Atkinson Hyperlegible; Lucide icons	Designed for low vision; every icon has a label

Table 3: Technology stack.

3.2 Tracking the user

MediaPipe PoseLandmarker returns 33 body landmarks per frame, both in the image and as approximate 3D "world" coordinates. Up to three people are detected and a **target lock** follows the user by the nearest torso position, so a passer-by is ignored. A **One-Euro filter** removes jitter without noticeable lag (it smooths strongly when the hand is still and lightly when it moves fast). Pose inference took a median of **44 ms** per frame on our laptop (full model); a lighter model is available for slow computers. Voice commands act on the recogniser's first partial result rather than waiting for the final one, which took 1–3 s in our pilot; their end-to-end delay will be measured in the study. Spoken feedback uses local system voices and starts immediately.

3.3 Fitting one exercise to every person

Arm length. The user holds the arm out to the side for three seconds. Upper-arm length is the shoulder–elbow distance in the image, forearm length the elbow–wrist distance, and arm length L their sum. A bone can only *look* shorter than it is (when it turns toward the camera), never longer, so we take a high percentile (85th) of each length rather than the median. During play the lengths are raised if the arm is consistently seen longer.

Body units. Paths are stored relative to the moving joint, in units of the user's own segment length. For the arm the origin O is the shoulder, x points outward along the shoulder line (mirrored for the left arm) and y points up. A wrist position P is stored and replayed as

$$p = R\,(P - O)\,/\,L, \qquad P = O_{user} + L_{user}\,R^{T}\,p.$$

The point (1, 0) therefore always means "a full arm's length straight out at shoulder height". A therapist whose wrist ends at 0.92 arm lengths produces a tape that ends at 0.92 of *each* user's arm length. The wrist and shrug exercises use the same idea with hand length and shoulder height.

Distance. Between strokes the current shoulder width is compared with the calibrated one and all lengths are rescaled slowly; the scale is frozen during a stroke so the tape never moves under the brush. If the user's size in the image stays outside 70–145% of the calibrated size for 1.5 s, the game pauses and asks them to return to their spot.

3.4 Depth from a single webcam

Forward raises and elbow bends move toward the camera, where a flat image under-reports them (a 60° forward raise looks like 0° from the front). Because the camera is far away compared with the arm's depth range, a segment of calibrated length L_0 that appears with length l must have a depth component

$$\Delta z = \sqrt{(\max(0,\ L_0^2 - l^2))},$$

whose sign (toward or away) comes from MediaPipe's world coordinates. Chaining shoulder, elbow and wrist gives 3D joint angles. In unit tests a 60° forward raise reads 60° in 3D and 0° in 2D. When a bone points almost straight at the camera the estimate is marked unreliable. Depth also drives an in-game cue: the brush grows as the hand comes forward. Depth cameras and monocular depth networks were rejected as unavailable at home or too slow in a browser.

3.5 From a therapist's video to an exercise

In Expert mode a therapist uploads a front-facing video of 3–5 slow repetitions; no photos or measurements are needed. The app then:

1. runs pose tracking on the video (15 frames per second) and picks the arm that moves more;
2. calibrates from the video itself (the arm passes flat to the camera at some point);
3. converts every frame to body units and 3D angles;
4. chooses the main movement with a simple rule (shoulder lift, elbow bend or sideways swing, whichever varies most); no other AI model is used;
5. splits the movement into repetitions, drops outliers, and averages the rest into one path.

The therapist sees the tracked path over the video and an angle chart, can exclude repetitions, sets the tape width and number of strokes, names the exercise and approves it.

3.6 Scoring a stroke

A stroke starts when the brush rests on the start circle for 0.6 s. The brush is then projected onto the tape and progress can only move forward; paint is laid only inside the tape. The stroke completes at 95% progress plus any end hold. Errors restart only the current stroke, never the wall (Table 4). **Neatness** (0–100) compares the performed and reference paths with Dynamic Time Warping. **Pay** is Rs 40 per stroke, a neatness bonus up to Rs 20 and Rs 50 per wall; it is never deducted.

Situation	Rule	Response
Brush off the tape	more than 1 s	Stroke restarts with a calm spoken line
Leaning the trunk sideways	more than 15°	"Keep your back straight"; stroke restarts
Hiking the shoulder	small / large	Spoken hint first; only a large hike restarts
User or wrist not visible	tracking lost	Auto-pause; never counted as the user's error
Repeated misses	2 misses / 3 neat strokes	Tape widens by 20% / narrows back

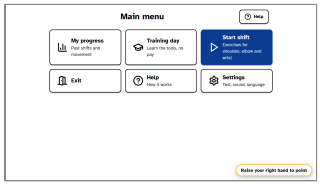
Table 4: Error rules. All were loosened after our pilot (Section 6).

4 Prototype Description

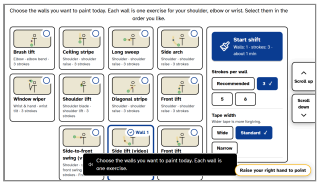
4.1 User journey

The full flow works end to end: **Main menu → Plan today's work → Setup check → Arm measurement → Training (first time) → Shift → Pay slip → My progress** (Figure 2). In *Plan today's work* the user or a carer chooses which walls to paint, how many strokes and the tape width, from eleven exercises for the shoulder, elbow and wrist. The *setup check* confirms, one line at a time, that the person is visible, at a good distance, centred, well lit and alone. During the *shift* the wall shows the tape, the brush and a faint camera image; after each stroke a chime plays and the earnings appear. *Pause* (button, Space key, voice, or automatic when tracking is lost) offers Resume, Measure again, Skip this wall and End shift. The

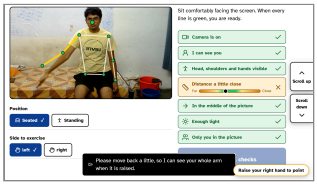
pay slip keeps the wage separate from a labelled movement summary (strokes, neatness, range of motion). Progress is saved after every stroke.



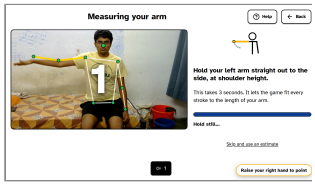
(a) Main menu



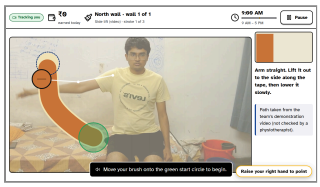
(b) Plan today's work



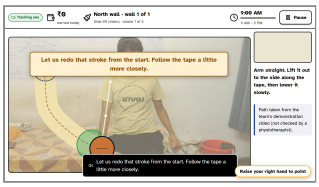
(c) Setup check



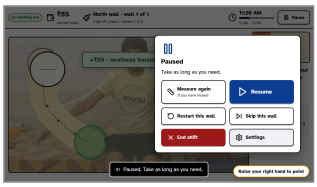
(d) Arm measurement



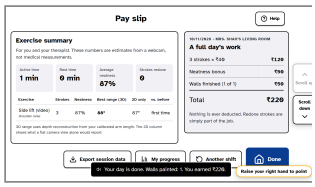
(e) Painting a stroke



(f) Stroke redone, tape widened



(g) Pause



(h) Pay slip

Figure 2: The Painter journey, captured from the running prototype.

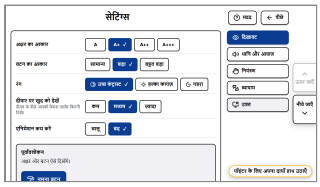
4.2 Interaction design for older adults

Table 5 summarises the HCI features and why each was chosen.

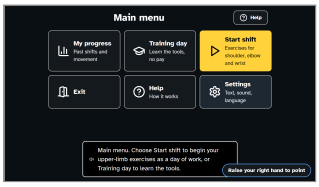
Feature	Design	Reason
Typeface	Atkinson Hyperlegible, 24 px by default (20–34 px)	Built to tell similar letters apart for low vision
Colour	High contrast (default, 21:1), warm paper, dark theme	Exceeds WCAG AAA; warm theme reduces glare
Redundancy	Every state shown with icon, text and colour, and spoken	Works with colour blindness and hearing loss
Targets	Buttons 72–108 px high, ≤6 per screen	Fitts' and Hick's laws; tremor tolerance
Hand pointer	The *free* hand points; the exercising hand paints	Menus never interrupt the exercise
Raise to point	Pointer appears only when the hand is raised	A hand resting on the lap never clicks
Dwell select	Hover fills a ring in 1.5 s (1–3 s)	No pinch or grip, which is often impaired
Reach box	Small box on the hand's own side maps to the whole screen	Small movements; no reaching across the body
Side layout	Main buttons mirror to the pointing hand's side	Shorter movements; large scroll buttons there too
Voice commands	pause, resume, repeat, next, back, help (English, Hindi)	Hands-free control; optional
Speech, captions	Every instruction spoken (rate 0.9, adjustable) and captioned	Low vision, low literacy, muted laptops
Earcons	Soft tones of 260–1100 Hz, under 0.35 s, one meaning each	Age-related hearing loss affects high pitches
Language	Full English and Hindi: text, speech and voice commands	Target users in India
Tone	“Let us redo that stroke”, never “wrong”; no confetti	Dignity and adult framing

Table 5: HCI features of the prototype.

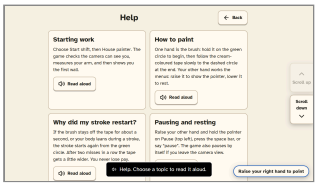
Mouse, touch and keyboard (Space to pause, R to repeat, Esc for back) work everywhere. All settings apply live and are saved (Figure 3). They cover text and button size, theme, language, voice and sound, captions, hand control, dwell time, exercising arm, seated or standing, tape width, posture checks, shift length and reduced motion.



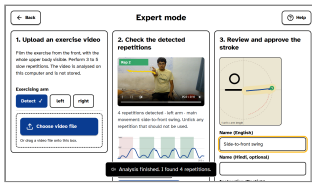
(a) Settings in Hindi



(b) Dark theme, large text



(c) Warm-paper theme



(d) Expert mode

Figure 3: Personalisation and the therapist's view.

5 Compromises in Prototyping

Dimension	Decision and justification
Vertical (deep)	The Painter trade end to end with real computer-vision scoring, all input and output modes, settings, two languages and saving. Depth was chosen over breadth because the main risk is whether webcam tracking can be fair and personal.
Horizontal (shallow)	Gardener, Postal sorter and Carpenter are shown as "coming later"; Help is static; no accounts, reminders, therapist dashboard or cloud sync.
Exercise content	Built-in paths follow standard exercise descriptions and team recordings; no physiotherapist has checked them yet. Expert mode lets a professional replace them.
Depth	Geometric estimate that assumes the user stays in place; approximate, not clinical-grade.
Movements	Shoulder, elbow and wrist. Finger and grip exercises need a hand-tracking model.
Voice input	Uses the browser's online speech service (Chrome and Edge); off by default.
Platform	Desktop browser with a webcam; one user per device; no clinical validation.
Wizard of Oz	None: everything in the demo is implemented.

Table 6: Prototyping compromises.

6 User Evaluation

6.1 Formative testing

Before the study the team tested the prototype on itself, and with scripted runs that feed recorded exercise videos to the browser as the webcam. Table 7 lists the main changes. In addition, 31 automated tests on synthetic skeletons check calibration, 3D angles, scaling, resets and voice matching.

Finding	Change
Using the exercising hand for menus broke both	Free-hand pointer with raise-to-point
Small shifts in the chair caused resets	Adaptive scale; posture judged per stroke; 1 s grace; tape widening
A fully stretched arm went past the end of the tape	High-percentile arm length, refined during play
Voice commands felt slow or did not respond	Act on partial results; ignore the game's own voice
Scrolling was impossible and reaching across the body tiring	Scroll buttons and layout on the hand's side
Users could not choose their exercises	"Plan today's work" and "Skip this wall"
The hand pointer worked only after a shift began	Camera starts on the main menu

Table 7: Formative findings and the design changes they led to.

6.2 Planned user study

> The study runs after this draft; results will be added here. Materials (consent form, script, tasks, questionnaire, logging sheets) are in `Evaluation_Data/`.

Participants. 5–8 people: ideally 2–3 adults aged 60+, the rest proxy users, and if possible one physiotherapist to review exercises in Expert mode. Proxies are reported separately.

Setting and tasks. A quiet room, a laptop webcam, the participant seated 1.5–2 m away, with concurrent think-aloud. Tasks: (T1) get ready (setup check); (T2) arm measurement; (T3) training; (T4) paint one wall; (T5) take a break and continue; (T6) make the text bigger; (T7) read the pay slip.

Measures. Task success and time; errors (wrong selections, tracking losses, resets); Single Ease Question per task; System Usability Scale; Borg CR10 effort before and after; three dignity items (e.g. "the game treated me like an adult"); preferred input method; short interview. Team members also run a heuristic evaluation (Nielsen's ten heuristics) on parts they did not build.

Accuracy and latency. Participants hold side and forward raises at about 45° and 90° while a phone goniometer measures the angle; we compare the app's 3D and 2D readings. Voice-command delay is timed from the session video.

Analysis. Descriptive statistics per task; SUS against the benchmark of 68; dignity scores; thematic analysis of comments; design changes ranked by severity.

Results [to be added]: participants; task success and time; SUS mean ± SD; SEQ; Borg; dignity scores; preferred input; angle error 3D vs 2D; voice delay; themes with quotes; changes made.

7 Ethics and Risk Assessment

Risk	Mitigation
Fatigue, pain, falls	Seated by default; mild exercises; short sessions; no time pressure; rests and lunch break; pause at any time; stop if pain, dizziness or Borg ≥ 5; a team member supervises; "Not a medical device" notice in the app
Privacy	Camera frames are processed on the device and never stored or sent; only an optional nickname, settings and numbers are saved; participants coded P1, P2, ...; no video or audio recorded
Voice data	Voice commands are off by default, and Settings explains that audio goes to the browser's speech service
Feeling judged	Adult wording; no deductions; participants told "we are testing the game, not you"
Clinical claims	Movement numbers labelled as estimates; no diagnosis or medical advice

Table 8: Risks and mitigations.

Consent. Each participant receives an information sheet and signs a consent form (English or Hindi). *Participation is voluntary; participants may stop or withdraw at any time without giving a reason, and no sensitive personal or health data is collected.*

8 Conclusion

The prototype shows that the Assignment 2 concept works on ordinary hardware. One webcam and in-browser pose tracking are enough to fit one reference movement to every user, recover movement toward the camera, and score strokes in a forgiving way. A therapist's phone video becomes a playable exercise without measurements or extra models. Around this core, the interface applies age-appropriate design: readable type and contrast, large dwell targets for the free hand, spoken and captioned feedback, voice and keyboard alternatives, Hindi, and a penalty-free adult metaphor. The main limitations are that no physiotherapist has validated the exercises and that depth is approximate. Next steps are the user study with older adults, hand tracking for finger exercises, more trades and a therapist dashboard.

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